Design and Development of a Wall Traversing Robot for Inspection Purposes

DAB REPORT

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Abstract

The purpose of this project primarily deals with the design and development of a robot that can traverse a ceiling or a vertical plane at a considerable height. Negative thrust based suction will be used to hold the robot against the vertical surface and 4-wheel motor combination for locomotion. The robot can be controlled manually or can be automated to traverse in multiple directions across the surface and can be utilized for numerous applications, such as security inspection, glass wall cleaning, pattern drawing, safety monitoring, fire rescue, and videography from complicated angles. It will further reduce safety hazards for people working at elevated levels and in those areas where it is difficult for humans to reach.

1. Introduction

The advent of robotics as we know dates back to 20th century when George C. Devol, an inventor from Kentucky, patented a reprogrammable manipulator named “Unimate”. This is however a contraption from a long time ago. Presently, robotics is a vast field which has applications in virtually every aspect of life. From automating processes in the industry to creating machines for household tasks, robots have become the norm.

The major reasons for the usage of robots is improvement in productivity, efficiency and reduction of manual work. This also involves enhancing the safety and protection of human life. For instance, construction at high rise buildings puts human life in jeopardy. A 2006 Bureau of Labor Statistics reported 1226 deaths in construction workplaces, out of which 809 were a result of falls from raised structures. Such instances open up a window of demand for development of robots which can take over manual workforce and reduce the troubling safety hazards.

Operations at elevated surfaces might prefer the use of drone technology but, drones and quadcopters are confined to tasks where no surface contact is required. To fill this void, development of wall climbing robots was worked upon, with different mechanisms for locomotion on vertical surface. Different procedures of adhesion, suction and thrust were tested to attain competent outcomes.

This project will be employing the method of negative thrust by an Electric Duct Fan, with a gear-motor mechanism for locomotion. The goal of this project is to build a modular robot, which can traverse on complicated vertical and elevated surfaces where human intervention poses threats. It will act as a platform for various applications where related mechanisms for the role can be integrated with the base model. Applications such as surveillance, emergency rescue, cleaning, pattern illustration and image processing can be incorporated with the module.

1.1. Problem Statement

The objective of this project is to design, fabricate and develop a modular robot which can traverse on vertical and elevated surfaces with an effective communication system enabling it to be controlled manually or traverse autonomously. This project aims to aid in improving safety of people working at heights and other dangerous sites by removing the need of manual work in respective areas.

1.2. Design Concept

In order to achieve this objective, the project has been divided into a list of landmarks that have to be hit in order to obtain the final product. The list of landmarks is as follows:

1. Develop a modular, open source design that can be implemented for various applications.
2. Fabricate a model using Solidworks which is mechanically stable and efficient.
3. Analyze the kinematics for thrust and locomotion of the robot.
4. Decide the model of EDF for negative thrust.
5. Determine the robot's orientation with the use of sensors.
6. Design and implement a remote-control system
7. Design and implement a method for autonomous movement.
8. Integration of on-board Computer for various applications.

1.3. Literature Review

The research and development of robots which can climb and move on vertical surfaces has been in progress, with different mechanisms and methods to achieve their tasks. We can go back a bit far as 1994 when research was carried out for robots which can inspect nuclear plants.

In recent past, various other designs were tested and implemented in order to achieve a light weighted, cost efficient and a reliable robot. Robots based on adhesion, suction and pneumatic principles were developed to achieve the task.
Majority of the research is done the design rather than actual realization of the robot.

Year	Research Paper	Authors
1994	Robicen: A wall-climbing pneumatic robot for inspection in nuclear power plants	Leoncio Briones, Paul Bustamante and Miguel A.Serna[§]

1994	Wall-climbing robot for inspection in nuclear power plants	L. Briones , P. Bustamante , M.A. Serna
2002	A low-cost lightweight climbing robot for the inspection of vertical surfaces	Guido La Rosa Michele Messina, Giovanni Muscato and Rosario Sinatra
2011	Wall Climbing Robots with Track-wheel Mechanism	Giuk Lee, Junhwan Park, Hwang Kim, Kunchan Seo, Jongwon Kim and TaeWon Seo
2011	Wall climbing robot: Mechanical design and implementation	Abdul gani Albagul, A Asseni and Othman O. Khalifa
2013	Climbing Robots for Maintenance and Inspection of Vertical Structures	Daniel Schmidt and Karsten Berns
2014	A Low-Cost, Light-Weight Climbing Robot for Inspection of Class Curtains	Ran Liang, Meteb Altaf, Eball Ahmad and Rong Liu, Ke Wang
2015	Wall Climbing Robot for Inspection of Wall Using Digital Image Processing	Behera, Abyarth Kumar
2015	Development of a Wall Climbing Inspection Robot with High Mobility on Complex Shaped Walls	Yuan Chang
2016	A survey on pneumatic wall-climbing robots for inspection	Angellica Brussel, George Andrikopoulos and George Nikolakopoulos
2016	Wall Climbing Robot for Dust Cleaning in a High Risk Buildings	R. Geethanjali, Mr.M.Sudhakaran, Dr.R. Seyezhai

2017	Wall Climbing Robot based on Suction Technique	Khyati Meena, Shubham Gupta and Vijay Khare
2017	Tele-operated propeller-type climbing robot for inspection of petrochemical vessel	Alkalla, M., Fanni, M., Mohamed, A. and Hashimoto, S.

2. Design & Methodology

In most of the aforementioned studies, a considerable amount of money and time has been spent in the design of the robot. They are expensive in their fabrication and complex in their control mechanisms. The aim of this project is to present a modular, scalable design that can be used as a platform for any wall traversing robot. This will allow upcoming researchers to spend their time and money on locomotion/automation algorithm improvements instead of robot design and fabrication.
The robot has thus been divided into two workspaces, the Mechanical structure of the robot and the Electronics.
The mechanical structure consists of complete design and fabrication of the body. The electronic structure can be divided into two subsections: Control system and communication system. The duty of the control system is to manage the thrust, locomotion and actuation of the robot. The communication system has to achieve proper connection from the outside devices such as remote and path planner.

2.1. Mechanical Design

The mechanical design this project is primarily divided into two parts:

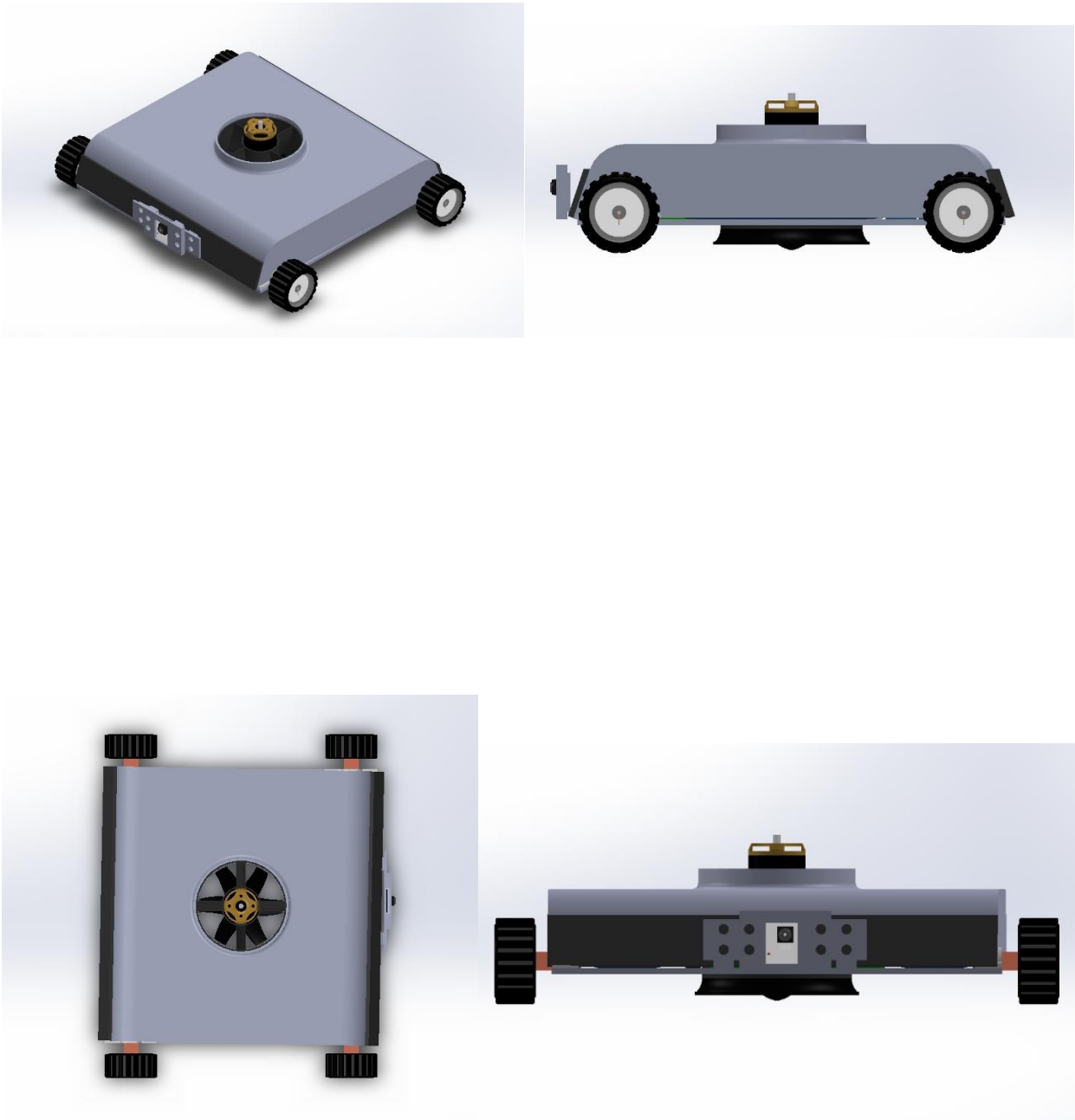
1. Canopy: This will be the upper part and will be 3D printed using materials such as ABS or PLA.

2. Base Plate Chassis: This will be the lower part which will be made of acrylic. All the components will be mounted on this structure.

We place the Electric Duct Fan at our assumed Centre of Gravity (CG) point of the body. The rest of the components are aligned with CG's lateral axis. The Base Plate Chassis is chosen as to lower the components of the robot, as the CG of the robot will be effected if their positions are changed. Nyloc nuts have been used to reduce the vibrations.

According to model analysis, brackets and mounts are used for effective maintenance and will nyloc nuts will damp the vibrations.

A cockpit structure will be used for integration which will create a cavity, which will be helpful in harnessing. Overall, the robot has a symmetric aerodynamic shape which will reduce drag.



Figure(s) 1: 3D Model of Robot

2.2. Electrical Schematic

Main control unit of our robot is the NodeMCU. It will control the movement of the gear motors and EDF, via ESC, based on the input it received from the controlling device (laptop, mobile). The sensors will constantly input their data into the NodeMCU which will process it accordingly. One such data would be the acceleration which was later processed to know the position of the robot. Raspberry pi is an additional computing unit for multiple applications in the future. We are using it to detect the images.

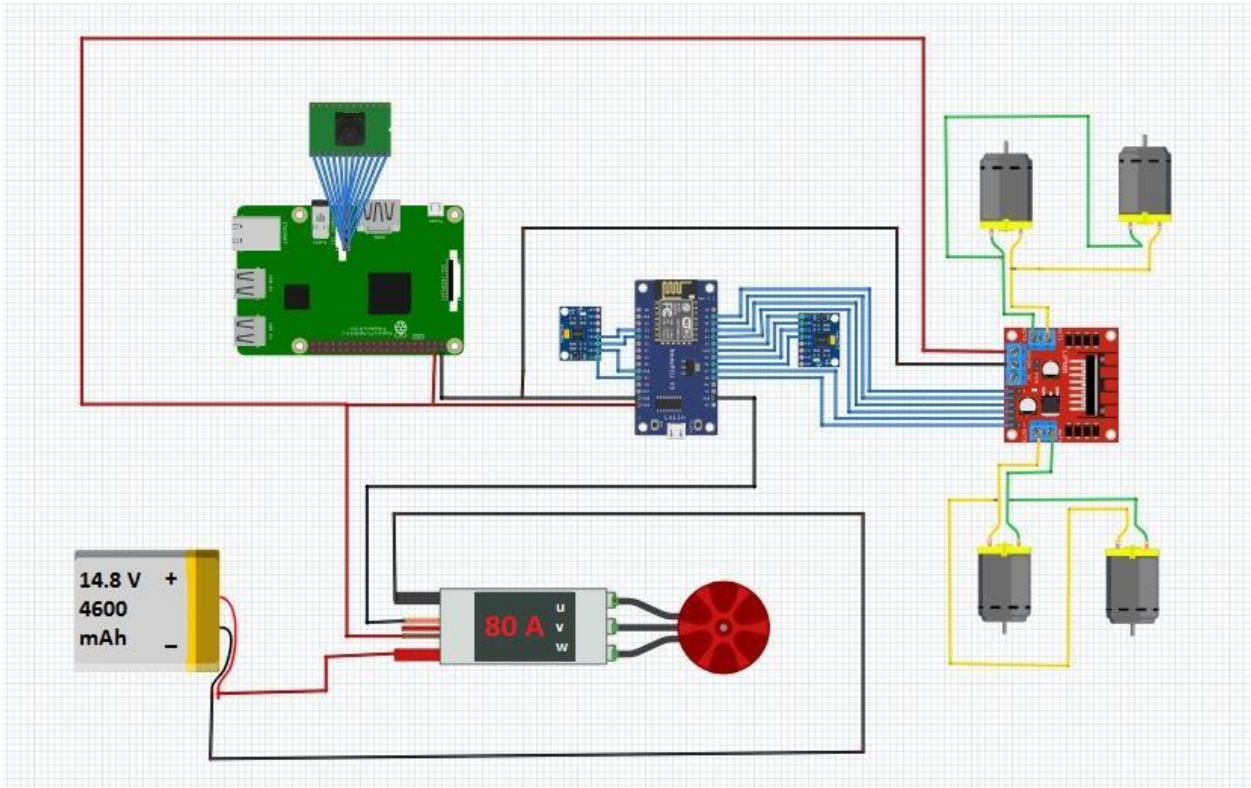


Figure 2: Schematic Diagram

3. Kinematics Analysis

3.1. Locomotion

Our wall traversing robot is motor driven by 4 hub mounted motors on each of its four wheels. The rotation of these motors will control the direction in which it is traversing. If all the motors rotate in the same direction, the robot will move in that direction. A differential between the right and left motors will steer the robot in its respective direction.

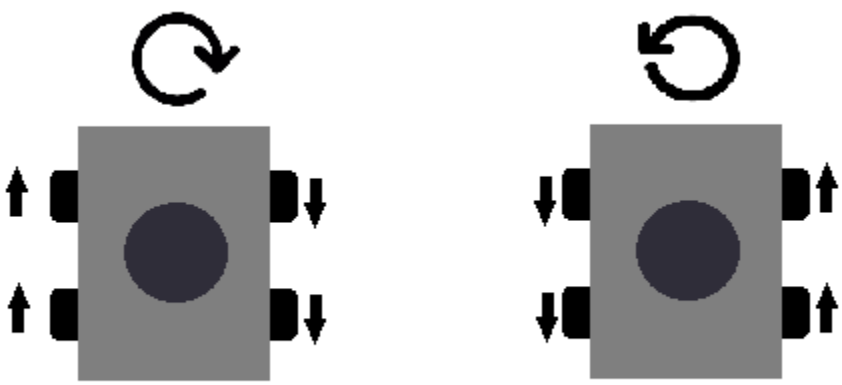


Figure 3: Illustration of Locomotion Mechanism

3.2. Adhesion

The robot's wall clinging mechanism is based on negative thrust exerted by the EDF unit. Negative thrust will push the robot against the wall and in turn cling it to the wall. The minimum force needed to push the robot is given by the formula:

$$F = \frac{m.g}{\mu}$$

where F is the force exerted on the wall

m is the mass of the robot

g is the gravitational pull exerted on to the robot

μ is the coefficient of friction between the tyres of the robot and the material of the wall.

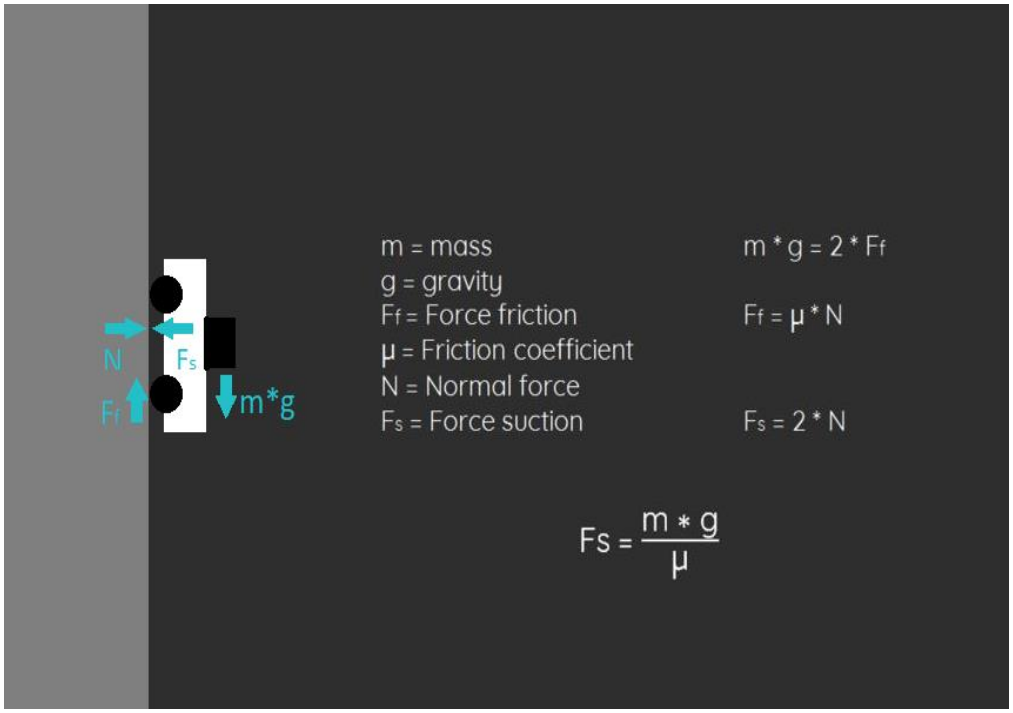


Figure 4: Forces acting on robot & formula involving them

3.3. Positioning

The position of the robot will be determined through the measurement of the RPM of the gear motors. The encoders installed with the gear motors will measure its RPM at each instant. We will derive the distance covered by the robot from the RPM by using the following approach:

1. Distance covered in one complete rotation of the wheel: 2πr; where r = radius of the Wheel.
2. Distance covered in a minute: RPM * D; where D = 2πr.

We will also use an MPU 6050 to measure its angle from a reference line. Using this angle we may be able to know the position of the robot in the (x, y) coordinates at each instant.

Y = S*sin(α);

X = S*cos(α);
where S = distance covered

α = angle from the base line

To further ensure that the position is accurate we will double integrate the acceleration measured from the MPU 6050 to calculate the distance. We will average out the distances from both the methods to get a more accurate result.

S = $\iint A \ dt$: where S= distance covered

A= acceleration

t= time for integral

4. Design Components

4.1. Electronic Ducted Fan Unit

The method of negative thrust will be used in this robot for clinging to the vertical surfaces. An Electric Ducted Fan (EDF) is selected based on calculations to provide the best thrust to weight ratio. The EDF is chosen by analyzing its thrust and comparing it with the expected mass of the robot. An ESC is also selected alongside which controls the required current for an EDF. This robot will be using a QX-Motor 70mm 6 Blade EDF unit with 2300KV Brushless Motor.

Figure 5: EDF Unit with Motor

Specifications of 3500KV Motor:

KV	3500	Idle Current 10.0V	1.3A
Configuration	12N8P	No. of Cells (Lipo)	3-4S
Stator Diameter	22mm	Max Continuous Current(A)/60s	70A
Stator Length	27mm	Max Continuous Power(W)/60s	1000W
Shaft Diameter	4mm	Max Efficiency	>70.5%
Motor Dimension	28x48mm	Internal Resistance	0.012ohms
Weight	107g	Recom. ESC	85A

Table 1: BL Motor Specifications

4.2. Microcontroller:

The control functions will be operated using the microcontroller so should have all the means for communication with EDF, gear motors and sensors. NodeMCU was selected on the basis of its features which fulfills our requirements.

4.2.1. NodeMCU

NodeMCU is based on the Esperessif ESP8266-12E WiFi System-On-Chip, loaded with an open-source, Lua-based firmware. it's perfect for IoT applications, and other situations where wireless connectivity is required. This chip has a great deal in common with the Arduino – they're both microcontroller-equipped prototyping boards which can be programmed using the Arduino IDE.A brief detail about NodeMCU:

NodeMCU comes with a good spaced of 128KB RAM

It has 4MBytes of ROM (flash).

It comes with micro USB port which is readily available.

NodeMCU development board is smaller in size compared to Arduino UNO.

It has almost the same price as other controllers but with additional features.

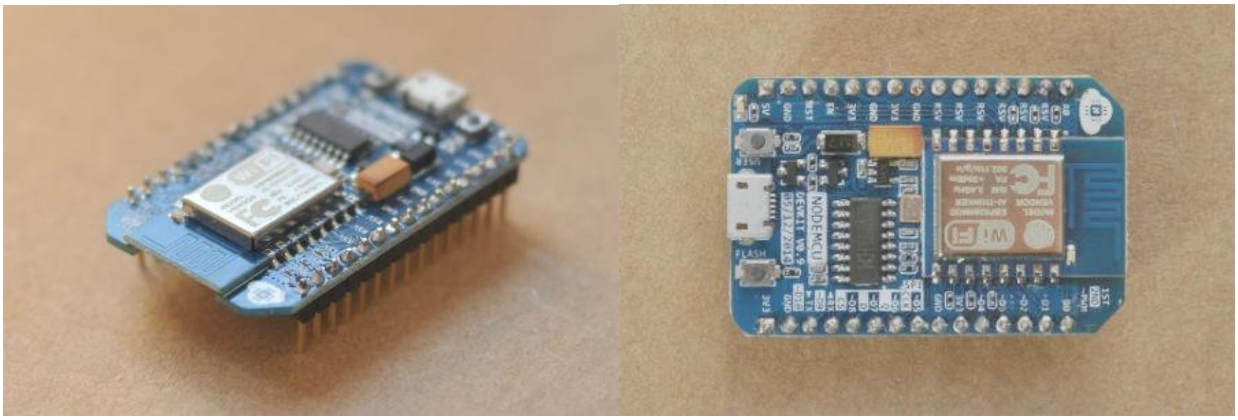


Figure 6: NodeMCU

4.2.2. Power Requirement

As the operating voltage range of ESP8266 is **3V to 3.6V**, the board comes with a LDO voltage regulator to keep the voltage steady at 3.3V. It can reliably supply up to 600mA, which should be more than enough when ESP8266 pulls as much as **80mA during RF transmissions**. The output of the regulator is also broken out to one of the sides of the board and labeled as 3V3. This pin can be used to supply power to external components.

4.2.3. Peripherals and I/O

The ESP8266 NodeMCU has total **17 GPIO pins** broken out to the pin headers on both sides of the development board. These pins can be assigned to all sorts of peripheral duties, including:

- **ADC channel** – A 10-bit ADC channel.
- **UART interface** – UART interface is used to load code serially.
- **PWM outputs** – PWM pins for dimming LEDs or controlling motors.
- **SPI, I2C & I2S interface** – SPI and I2C interface to hook up all sorts of sensors and peripherals.
- **I2S interface** – I2S interface if you want to add sound to your project.

4.2.4. Serial Communication

The board includes CP2102 USB-to-UART Bridge Controller from Silicon Labs, which converts USB signal to serial and allows your computer to program and communicate with the ESP8266 chip.

- CP2102 USB-to-UART converter
- 4.5 Mbps communication speed
- Flow Control support

4.2.5. ESP8266 NodeMCU Pinout

The ESP8266 NodeMCU has total 30 pins that interface it to the outside world. The connections are as follows:

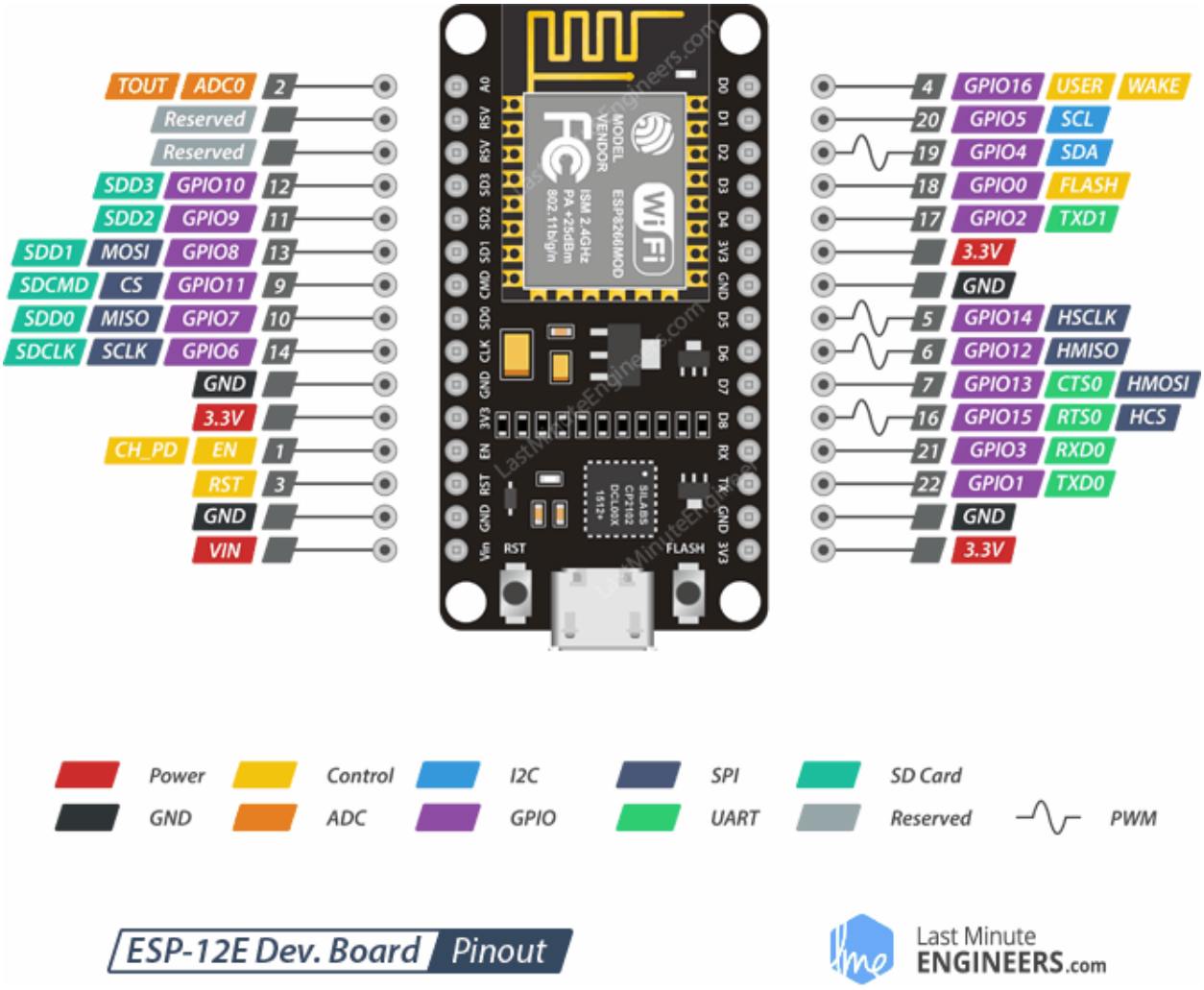


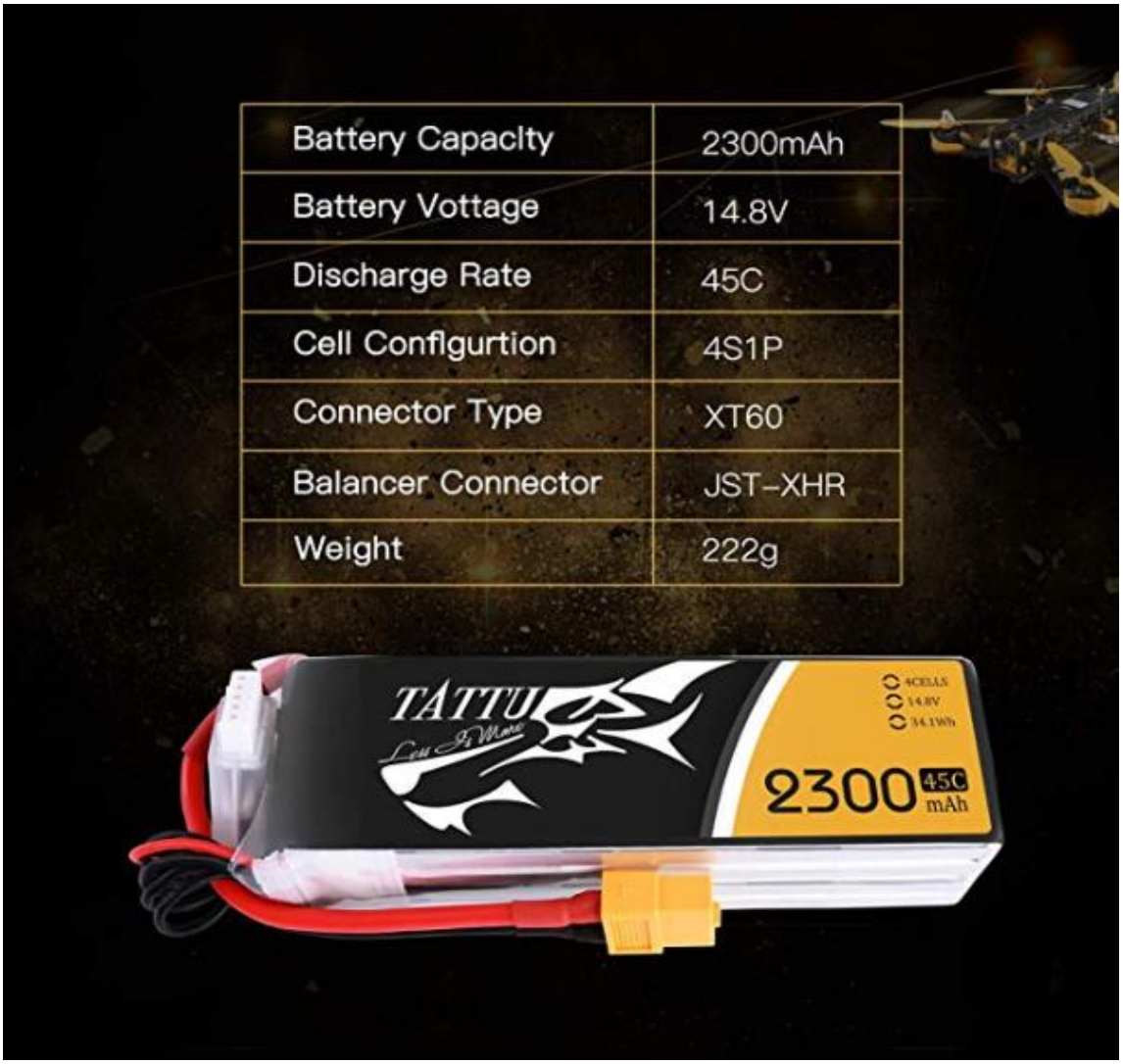
Figure 7: Pin Diagram of NodeMCU

4.3. Batteries:

Since the robot will be traversing along complicated surfaces for various applications, it needs a considerable working time for operation. Thus, battery selection is an important process which directly controls the operating time of the robot.

The batteries will be providing approximately 5-6 minutes of operation which is sufficient for the relevant tasks for the robot. The batteries are carefully selected so they do not contribute majorly to the total weight of the robot and can provide maximum time period. A couple of 2300mAh batteries will be used in this project.

Table 2: Battery Specifications



Battery Capacity	2300mAh
Battery Voltage	14.8V
Discharge Rate	45C
Cell Configurtion	4S1P
Connector Type	XT60
Balancer Connector	JST-XHR
Weight	222g

Figure 8: Lipo Battery

4.4. Electronic Speed Controller

An ESC is integrated with the EDF Unit in order to provide adequate current to the ducted fan for the required thrust by the user. The speed of the EDF varies in relation to changes of thrust requirement.

Model	Readytosky 80A ESC
Burst Current	100A
Constant Current	80 A
BEC	5A/5V
Suitable Batteries	2~6S LiPoly
Throttle signal refresh rate (Hz)	50 ~ 432
Maximum speed	210000 2-pole motor rev / min, 6-pole motors 70,000 rev / min, 12 pole motors 35,000 rev / min
Length (mm)	9
Width (mm)	40
Height (mm)	13
Weight (gm)	100

Table 3: ESC Specifications

4.5. RASPBERRY PI 3 Model B+

- Raspberry PI will be integrated with video streaming to perform inspection applications. It has the following features:
- Broadcom BCM2837B0, Cortex-A53 (ARMv8) 64-bit SoC @ 1.4GHz
- 1GB LPDDR2 SDRAM
- 2.4GHz and 5GHz IEEE 802.11.b/g/n/ac wireless LAN, Bluetooth 4.2, BLE
- Gigabit Ethernet over USB 2.0 (maximum throughput 300 Mbps)
- Extended 40-pin GPIO header
- Full-size HDMI
- 4 USB 2.0 ports
- CSI camera port for connecting a Raspberry Pi camera
- DSI display port for connecting a Raspberry Pi touchscreen display
- 4-pole stereo output and composite video port
- Micro SD port for loading your operating system and storing data
- 5V/2.5A DC power input
- Power-over-Ethernet (PoE) support (requires separate PoE HAT)

Figure 9: Raspberry Pi 3 Model B+

4.6. Motor Driver

The L298N Motor drivers were employed.

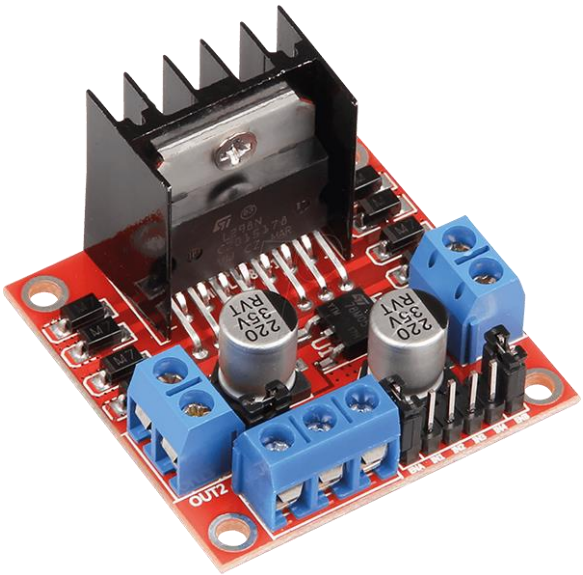


Figure 10: L298N Motor Driver

Chip	L298
Logic Voltage	5V
Drive voltage	5V – 35V
Logic Current	0mA – 36mA
Drive Current	2A
Storage Temperature	-20C to 135C
Max Power	25 W
Weight	30 g
Size	43mm*43mm*27mm

Table 4: L298N Specifications

4.7. Gear Motors

Light weight, high torque and low RPM. Fine craftsmanship, durable, not easy to wear. With excellent stall characteristics, can climb hills easily. You can also easily mount a wheel on the motor's output shaft. Widely used on boat, mode l car, robotic, home appliances, linear motion control.

Model: GA12-N20

•Rated Voltage : 6~12V
•Revolving Speed : 100RPM @ 6V
•Load Speed: 80RPM
•Rated Torque: 2 kg.cm
•Stall Torque: 16 kg.cm
•Rated Current: 0.07A•
Stall Current: 1A
•Reduction Ratio: 1:10
•Total Length : 34mm
•Gear Material: Full Metal
•Gearbox Size : 15 x 12 x 10mm (L*W*H)
•Shaft Size : 3 x 10 mm(D*L)
•Net Weight : 10g

4.8. Transmitter

This project will be using FlySky FS–i6 Transmitter. Its specifications are:
Frequency: 2.4GHz ISM Frequency Range.

- Modulation: GFSK.
- Spread Spectrum Mode: AFHDS 2A and AFHDS.
- Band: 142.
- Number of Frequency Channels: 20.

- Output Power: <=20dbm.
- Working Current: <=100mA.
- Working Voltage: 1.5v x 4 AA

Figure 11: Transmitter

4.9. Receiver

The FS-i6 receiver specifications are as follows:
Channel: 6.

- Frequency Range: 2.4055–2.475 GHz.
- Band Width Number: 140
- Transmitting Power: ≤ 20 dBm.
- RF Receiver Sensitivity: 105 dBm.
- 2.4G Mode: The second generation of an enhanced version of the automatic FM digital system.
- Encoding: GFSK.
- Antenna Length : 2 x 26 mm (dual antenna).
- Input Power : 4.0 – 8.4 VDC (2A).
- Dimension : 47 x 26.2 x 15 mm
- Weight: 14.9g

Figure 12: Receiver

5. Algorithms

5.1. Path Following

To control the robot for path following, the Processing IDE will be utilized. Processing is a great source for creating graphics, illustrations, etc.

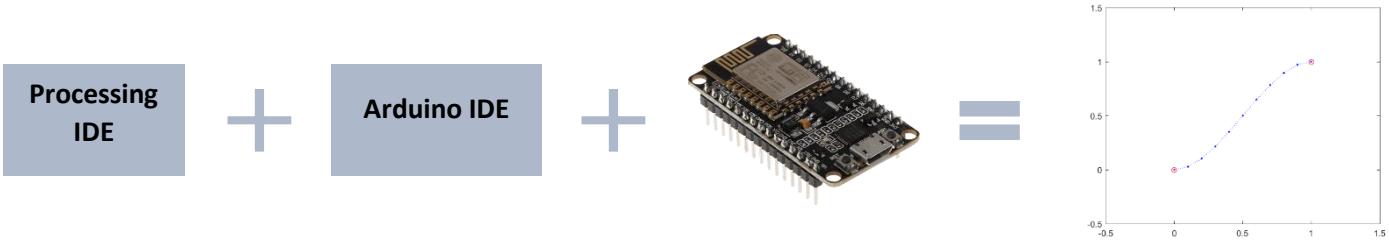


Figure 8: Flowchart of Trajectory Mapping

The processing IDE will communicate with the Arduino IDE. In the Arduino IDE, the code will run which sends commands to the ESP826 chip on the NodeMCU. This is how to control it from the laptop using Wi-Fi.

Using Processing IDE, a program is created which provides the user with a graphical interface. On it, the user can conceptualize a path by selecting a series of points. These points will be joined and illustrated into a path by the IDE. The Processing IDE will send those details to the Arduino IDE, which sends them further to the NodeMCU. The NodeMCU will control the robot and make it move along the path provided by the user.

This algorithm can be utilized in multiple applications such as:

- Pattern drawing
- Sketching
- Area Mapping

5.2. Crack Detection

With the advancement in computer vision algorithm, researchers have been attracted to use deep learning algorithm to solve the structural damage detection autonomously. Deep learning is a part of machine learning which has multiple layers of nonlinear units for transformation as well as feature extraction and each of the successive layers uses the output of the previous result as the layer input. It can be unsupervised, semi-supervised, and supervised learning. Nowadays, many companies use diverse deep learning architectures for computer vision, drug design, medical image analysis, speech recognition, audio recognition, social network filtering, as well as programming challenging games. In this research works, we are deploying supervised deep learning specially convolution neural network to detect several types of flexible and rigid pavement cracks.

We shall be using Mobilenet as it is lightweight in its architecture. It uses depth-wise separable convolutions which basically means it performs a single convolution on each color channel rather than combining all three and flattening it. MobileNet model was developed at Google, pre-trained on the ImageNet dataset with 1.4M images and 1000 classes of web images. We will use this as our base model to train with our dataset and classify the images of cats and dogs.

The model was trained on the Concrete Crack Images for Classification from Medeley. It contains concrete images having cracks. The data is collected from various METU Campus Buildings.

The dataset is divided into two as negative and positive crack images for image classification.

Each class has 20000images with a total of 40000 images with 227 x 227 pixels with RGB channels.

The dataset is generated from 458 high-resolution images (4032x3024 pixel) with the method proposed by Zhang et al (2016).

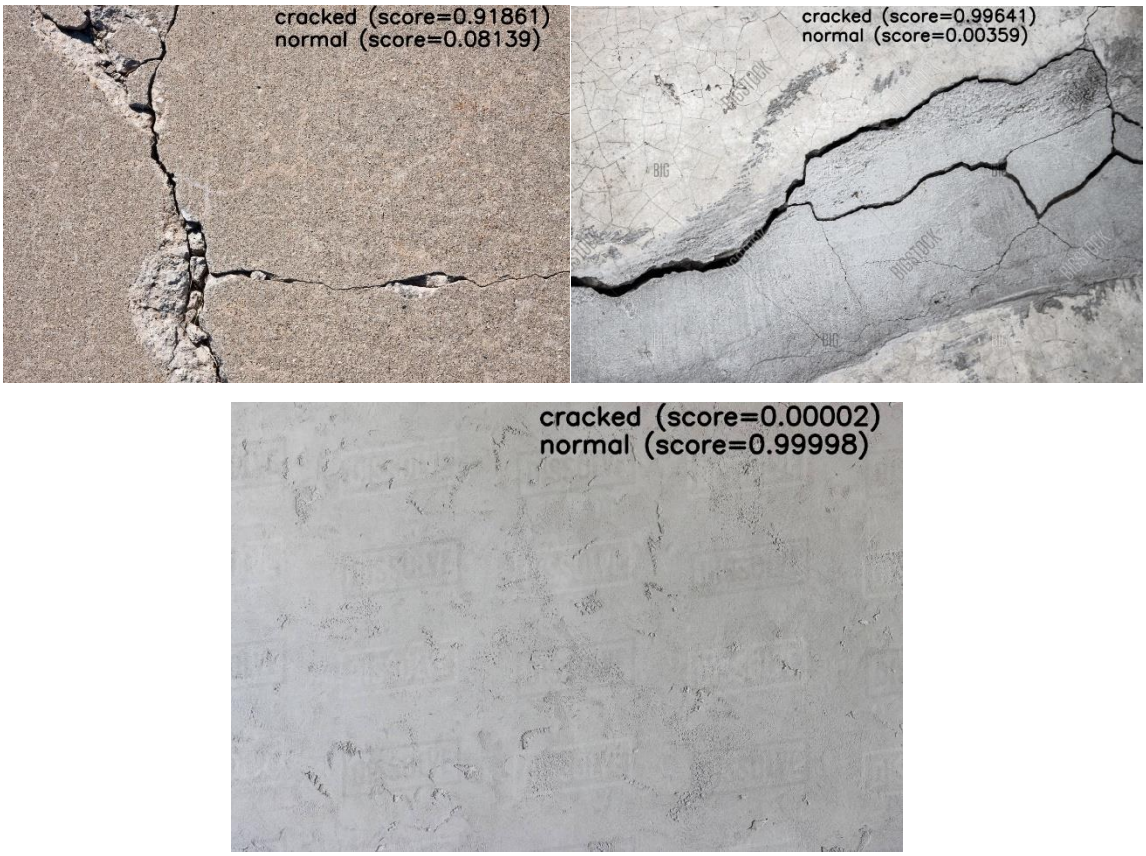


Figure 14: Results of Crack Detection Algorithm

6. Conclusion

The applications of wall traversing robots can be numerous as their nature of work is dangerous if done by a human. These robots can perform their jobs at heights and also at locations which has safety concerns such as:

- Surveillance and security inspection at towers, in industries etc.
- Pattern drawing and painting on ships, planes etc.
- Cleaning of high-rise glass buildings.
- Inspection in places which has radiation hazards such as nuclear plants.
- Fire rescue operations
- Crack detection in pipelines, walls, wind turbine etc.

The major advantage of a robot in dangerous environments is that it doesn't necessarily have to come back. Its existence pales in comparison to a human life, thus creating a vast array of opportunities for its use.